

PneumAI Database Schema

PostgreSQL Database Structure | Generated: December 11, 2025

Database Name	pneumai_db
Database Type	PostgreSQL 14+
Total Tables	7
Character Set	UTF-8

PATIENTS

Stores patient information and medical records

Column Name	Data Type	Constraints	Description
id	VARCHAR(50)	PRIMARY KEY	Unique patient identifier
name	VARCHAR(255)	NOT NULL	Patient full name
email	VARCHAR(255)	UNIQUE, NOT NULL	Patient email address
phone	VARCHAR(50)		Contact phone number
date_of_birth	DATE		Patient date of birth
gender	VARCHAR(20)		Patient gender
medical_history	TEXT		Medical history notes
created_at	TIMESTAMP	DEFAULT NOW()	Record creation time
updated_at	TIMESTAMP	DEFAULT NOW()	Last update time

total_scans	INTEGER	DEFAULT 0	Total number of scans
last_visit	TIMESTAMP		Last visit timestamp

DOCTORS

Healthcare professional information

Column Name	Data Type	Constraints	Description
id	VARCHAR(50)	PRIMARY KEY	Unique doctor identifier
name	VARCHAR(255)	NOT NULL	Doctor full name
email	VARCHAR(255)	UNIQUE, NOT NULL	Doctor email address
phone	VARCHAR(50)		Contact phone number
specialization	VARCHAR(100)		Medical specialization
license_number	VARCHAR(100)		Medical license number
created_at	TIMESTAMP	DEFAULT NOW()	Record creation time
updated_at	TIMESTAMP	DEFAULT NOW()	Last update time

SCANS

CT scan metadata and analysis results

Column Name	Data Type	Constraints	Description
id	VARCHAR(50)	PRIMARY KEY	Unique scan identifier
patient_id	VARCHAR(50)	FOREIGN KEY → patients(id)	Reference to patient
status	VARCHAR(50)	NOT NULL	Processing status

upload_time	TIMESTAMP	DEFAULT NOW()	Upload timestamp
processing_time	REAL		Processing duration (seconds)
detected	BOOLEAN		Cancer detected flag
confidence	REAL		Detection confidence (0-1)
risk_level	VARCHAR(20)		Risk level (none/low/medium/high)
top_class	VARCHAR(100)		Detected cancer type
file_size	INTEGER		File size in bytes
image_format	VARCHAR(20)		Image format (DICOM/JPEG/PNG)
image_width	INTEGER		Image width in pixels
image_height	INTEGER		Image height in pixels
created_at	TIMESTAMP	DEFAULT NOW()	Record creation time

DETECTIONS

Individual cancer detections from YOLOv12 analysis

Column Name	Data Type	Constraints	Description
id	SERIAL	PRIMARY KEY	Auto-increment ID
scan_id	VARCHAR(50)	FOREIGN KEY → scans(id)	Reference to scan
class_name	VARCHAR(100)		Cancer type detected
confidence	REAL		Detection confidence (0-1)
bbox_x	INTEGER		Bounding box X coordinate
bbox_y	INTEGER		Bounding box Y coordinate
bbox_width	INTEGER		Bounding box width
bbox_height	INTEGER		Bounding box height
size_mm	REAL		Lesion size in millimeters
shape	VARCHAR(50)		Lesion shape description
density	VARCHAR(50)		Lesion density description

APPOINTMENTS

Patient-doctor appointment scheduling

Column Name	Data Type	Constraints	Description
id	VARCHAR(50)	PRIMARY KEY	Unique appointment ID
patient_id	VARCHAR(50)	FOREIGN KEY → patients(id)	Reference to patient
doctor_id	VARCHAR(50)	FOREIGN KEY → doctors(id)	Reference to doctor

doctor_name	VARCHAR(255)		Doctor name (denormalized)
appointment_date	DATE	NOT NULL	Appointment date
appointment_time	TIME	NOT NULL	Appointment time
type	VARCHAR(100)		Appointment type
status	VARCHAR(50)	DEFAULT scheduled	Appointment status
notes	TEXT		Appointment notes
created_at	TIMESTAMP	DEFAULT NOW()	Record creation time
updated_at	TIMESTAMP	DEFAULT NOW()	Last update time

MESSAGES

Patient-doctor communication messages

Column Name	Data Type	Constraints	Description
id	VARCHAR(50)	PRIMARY KEY	Unique message ID
sender_id	VARCHAR(50)	NOT NULL	Sender user ID
sender_name	VARCHAR(255)	NOT NULL	Sender name
sender_role	VARCHAR(50)	NOT NULL	Sender role (patient/doctor)
receiver_id	VARCHAR(50)	NOT NULL	Receiver user ID
receiver_name	VARCHAR(255)	NOT NULL	Receiver name
content	TEXT	NOT NULL	Message content
read	BOOLEAN	DEFAULT FALSE	Read status flag
created_at	TIMESTAMP	DEFAULT NOW()	Message timestamp

SCAN_COMMENTS

Comments and feedback on scan results

Column Name	Data Type	Constraints	Description
id	SERIAL	PRIMARY KEY	Auto-increment ID
scan_id	VARCHAR(50)	FOREIGN KEY → scans(id)	Reference to scan
user_id	VARCHAR(100)	NOT NULL	Commenter user ID
user_role	VARCHAR(20)	NOT NULL	Commenter role
user_name	VARCHAR(100)	NOT NULL	Commenter name
comment_text	TEXT	NOT NULL	Comment content
parent_comment_id	INTEGER	FOREIGN KEY → scan_comments(id)	Parent comment (for replies)
created_at	TIMESTAMP	DEFAULT NOW()	Comment timestamp
updated_at	TIMESTAMP	DEFAULT NOW()	Last update time

Database Relationships

From Table	To Table	Relationship Type	Description
scans	patients	Many-to-One	Each scan belongs to one patient
detections	scans	Many-to-One	Each detection belongs to one scan
appointments	patients	Many-to-One	Each appointment has one patient
appointments	doctors	Many-to-One	Each appointment has one doctor
scan_comments	scans	Many-to-One	Each comment belongs to one scan
scan_comments	scan_comments	Self-referencing	Comments can reply to other comments

Database Indexes

Index Name	Table	Column(s)	Purpose
idx_scans_patient_id	scans	patient_id	Fast patient scan lookups
idx_scans_upload_time	scans	upload_time	Chronological scan queries
idx_appointments_patient_id	appointments	patient_id	Patient appointment lookups
idx_appointments_doctor_id	appointments	doctor_id	Doctor appointment lookups
idx_appointments_date	appointments	appointment_date	Date-based queries
idx_messages_sender_id	messages	sender_id	Sender message lookups

idx_messages_receiver_id	messages	receiver_id	Receiver message lookups
idx_detections_scan_id	detections	scan_id	Scan detection lookups
idx_scan_comments_scan_id	scan_comments	scan_id	Scan comment lookups